

Data Knowledge

BRD Create Table & Insert



Data Knowledge

What is a Database?

A database is an organized collection of data that can be easily accessed, managed, and updated. It stores information electronically in a structured format, often using tables.

Types of Databases:

- Relational Database: Stores data in tables (e.g., MySQL, PostgreSQL, Oracle, SQL Server).
- Non-relational Database: Stores data in formats like JSON, documents, or key-value pairs (e.g., MongoDB, Cassandra).

CREATE DATABASE database_name;

Definition of CREATE TABLE in SQL:

The CREATE TABLE statement in SQL is used to create a new table in a database. It defines the table name, its columns, data types, and any constraints (like primary key, not null, etc.).

```
CREATE TABLE table_name (  
column1 datatype [constraint],  
column2 datatype [constraint],  
...  
);
```

```
CREATE TABLE Employees (  
EmployeeID INT,  
FirstName VARCHAR(50),  
LastName VARCHAR(50),  
HireDate DATE,  
Salary DECIMAL(10,2)  
);
```

Data Type	Stores	Example
INT	Whole numbers	10, -100
FLOAT	Decimal numbers	5.5, -23.45
CHAR(n)	Fixed-length text	'A', 'Test '
VARCHAR(n)	Variable-length text	'Hello', 'Data'
DATE	Date only	'2023-08-15'
BOOLEAN	True/False	TRUE, FALSE
DATETIME	Date and time	'2023-08-15 10:45:00'

Definition of INSERT INTO

The INSERT INTO statement is used to add new records (rows) into an existing table.

Insert All Columns in Order

INSERT INTO table_name

VALUES (value1, value2, value3, ...);

Insert into Specific Columns

INSERT INTO table_name (column1, column2, ...)

VALUES (value1, value2, ...);

Questions:

1. Healthcare Domain

Question:

Write a SQL query to create a table named Patients with the following columns:

- patient_id (integer)
- name (varchar 100)
- gender (char 5)
- age (integer)
- admission_date (date)
- diagnosis (varchar 255)

2. Finance Domain

Question:

Create a table named Transactions with columns for:

- transaction_id (integer)
- account_number (varchar 20)
- amount (float)
- transaction_type (varchar 20)
- transaction_date (datetime)

3. Pharmaceutical Domain

Question:

Write a SQL query to create a table named Medicines that includes:

- medicine_id (integer)
- name (varchar 100)
- category (varchar 50)
- expiry_date (date)
- price (float)

4. Automobile Domain**Question:**

Create a table named Vehicles to store the following information:

- vehicle_id (integer)
- brand (varchar 50)
- model (varchar 50)
- manufacture_year (integer)
- price (float)
- fuel_type (varchar 20)

5. Education Domain**Question:**

Create a table called Students with these columns:

- student_id (integer)
- full_name (varchar 100)
- grade_level (varchar 10)
- enrollment_date (date)
- gpa (float)

6. E-commerce Domain**Question:**

Write a query to create a table named Orders with:

- order_id (integer)
- customer_name (varchar 100)
- product_name (varchar 100)
- quantity (integer)
- order_date (date)

- total_amount (float)

7. Write SQL INSERT INTO statements to insert 5 records into each of the above tables

1. Display all columns from the Patients table.
2. Show only transaction_id, amount, and transaction_date from the Transactions table.
3. Retrieve name and price from the Medicines table.
4. Display vehicle_id, brand, model, and fuel_type from the Vehicles table.
5. Show all columns from the Students table.
6. Retrieve order_id, customer_name, and total_amount from the Orders table.
7. Display name and diagnosis of all patients from the Patients table who are older than 50.
8. Show account_number and transaction_type for transactions where the amount is greater than 5000 from the Transactions table.
9. Retrieve name and expiry_date of all medicines that expire before '2026-01-01'.
10. Display full_name and gpa of students whose gpa is greater than 3.5 from the Students table.

Note

Please Note: When you provide your assignment to us please mention the below details or format.

Topic Name: Ex. Join

Date: 22/7/2025

Name:(Your name)

Share your assignment in Email id:dataknowledge.class@gmail.com

Format: PDF Question

no:

Ans: